Computer Graphics Laboratory Assignment Logsheet List (Unit-wise)

This **Lab Task List** document provides a total of **27 practical exercises**, organized chapter-wise to align with our lecture schedule and syllabus requirements. Individual labs will be formally assigned to you following the completion of each relevant class or at equivalent intervals.

Unit	Lab No.	Lab Title
Unit 2	Lab 1	Understanding & Implementing DDA Line Drawing Algorithm
	Lab 2	Understanding & Implementing Bresenham's Line Drawing Algorithm
	Lab 3	Understanding & Implementing Midpoint Circle Drawing Algorithm
	Lab 4	Understanding & Implementing Midpoint Ellipse Drawing Algorithm
	Lab 5	Understanding & Implementing Scan Line Polygon Fill Algorithm
	Lab 6	Understanding & Implementing Boundary Fill Algorithm
	Lab 7	Understanding & Implementing Flood Fill Algorithm
	Lab 8	Program for creating a simple car shape

Unit	Lab No.	Lab Title
Unit 3	Lab 9	Understanding & Implementing 2D Translation
	Lab 10	Understanding & Implementing 2D Rotation
	Lab 11	Understanding & Implementing 2D Scaling
	Lab 12	Understanding & implementing 2D Reflection
	Lab 13	Understanding & implementing 2D Shearing
Unit 4	Lab 14	Understanding & implementing 3D Translation
	Lab 15	Understanding & implementing 3D Rotation
Unit 3	Lab 16	Understanding & Implementing Window to Viewport Transformation
	Lab 17	Understanding & Implementing Cohen-Sutherland Line Clipping Algorithm

Unit	Lab No.	Lab Title
	Lab 18	Understanding & Implementing Liang Barsky Line Clipping Algorithm
	Lab 19	Understanding & Implementing Sutherland-Hodgeman Polygon Clipping Algorithm
Unit 4	Lab 20	Understanding & Implementing Parallel Projection in 3D
	Lab 21	Understanding & Implementing Perspective Projection in 3D
Unit 5	Lab 22	Understanding & Implementing Bezier Curve
Unit 9	Lab 23	Animation Basics: Windmill Rotation
Unit 10	Lab 24	Understanding & Implementing the basics of OpenGL
	Lab 25	Drawing a Lit Sphere using OpenGL
	Lab 26	OpenGL Project-1: Drawing a Color Cube & Spin using 3D Transformation

Unit	Lab No.	Lab Title
	Lab 27	OpenGL Project-2: Drawing a bouncing ball using OpenGL

Software Requirements & Configuration

1. All Laboratory assignments will be performed using CodeBlocks.
2. **Install and Configure CodeBlocks =>** https://docs.google.com/document/d/1wDQXOmXvt-WP_hpJfCtyGTYnpJh9Gtt1yjugEq7xBqo/edit?usp=sharing

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Lab Submission Guidelines

Please adhere strictly to the following requirements, as they are mandatory for successful completion of the practical component of this course:

1. Submission Deadlines:

- All lab reports must be submitted by the exact deadline specified within the individual **LAB Module** provided. Failure to submit on time will result in a **deduction of marks** in your final internal assessment.

2. Non-Submission Policy 🚫:

- Any lab that is not submitted will be marked as **0**.

- Crucially, **non-submission of required lab work will prohibit your attendance** in the corresponding final practical examinations.

3. Report Format & Structure:

- Each lab report must strictly follow the required structure and format that is explicitly detailed within its respective **Lab Module**.

4. Use of References :

- The "References" provided in the Lab Modules are intended solely as supporting documents or guides in case of confusion.
- Please note that submissions found to be entirely copied or plagiarized from the provided references will **not be checked** and will automatically receive a score of **0**.

5. Required Submission Mode ✍️🖨️:

- The required mode of submission is a **handwritten document**.
- The program output must be included in a **printed form only**.

With Regards,

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